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# Development and validation of an artificial intelligence-based model for cardiovascular disease prediction using longitudinal data

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## Abstract

**Background** This research uses machine learning (ML) models to determine significant predictive factors for CVD events in Iran, a country with high mortality rates. The study evaluates the effectiveness of deep learning and mixed-effects logistic models in predicting 10-year CVD incidence using longitudinal TLGS data.

**Methods** A total of 4,872 adults aged ≥ 30 years without a history of CVD at baseline (2006–2008) were followed until 2020. Following exclusions due to prevalent CVD or insufficient follow-up data, the final analytic sample comprised 1,942 men and 2,930 women. Baseline demographic, behavioral, and biochemical characteristics were utilized as input features. A clinical history examination was employed to identify and confirm incident CVD events, such as stroke and coronary heart disease (CHD). Using longitudinal data collected during a 10-year study period, deep learning models based on Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures were utilized to detect variations in risk factor dynamics. The area under the receiver operating characteristic curve (AUC) was employed to assess the model's performance.

**Results** During the follow-up period, 545 participants (11.2%) experienced CVD events. In both sexes, the GRU model outperformed the LSTM model. In women, the GRU achieved an AUC of 0.739, whereas in men it achieved 0.738, in both cases outperforming LSTM. Compared with the traditional mixed-effects model, discrimination was comparable in women (GRU 0.739 vs. 0.74) and higher in men (0.738 vs. 0.70). Even though the models were restricted to 21 commonly employed clinical variables, their performance was similar to that of larger-scale studies that included hundreds of variables.

**Conclusions** Deep learning models, such as GRU, can effectively leverage longitudinal medical data to predict future cardiovascular disease events. The models achieved performance comparable to studies that used far more variables, despite relying on a limited feature set.

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**Keywords** Prediction model, LSTM, GRU, CVD, Longitudinal data

## Background

A considerable global health risk is presented by non-communicable diseases (NCDs), primarily cardiovascular diseases (CVDs) [1, 2]. In Iran, CVD is the most prevalent cause of mortality, with incidence rates, mortality rates, and disability-adjusted life years (DALYs) exceeding global rates [3, 4]. Factors such as sedentary lifestyles, rising obesity rates, reduced physical activity, and an aging population are expected to increase the burden of cardiovascular diseases in the country [5]. Cardiovascular diseases are a leading global cause of illness and death, accounting for around 17.9 million deaths and 347.5 million DALYs in 2015 [6, 7]. While mortality from CVDs has declined in high-income countries, they remain a major cause of death in low- and middle-income nations, especially in the Eastern Mediterranean region [7]. In Iran, mortality patterns have shifted from infectious diseases to non-communicable diseases, with CVDs being particularly widespread [8]. This high prevalence poses serious health and economic burdens [9]. Even though mortality rates have dropped due to advances in healthcare, many patients are still experiencing long-term complications. Additionally, heart failure, recurrent heart attacks, or abrupt cardiac death may result from severe CVD [10, 11].

CVD is a chronic, complicated condition that is a significant global cause of long-term disability and premature death. Although surgical interventions and medications can be helpful in managing the disease, they do not heal it and can have negative effects on quality of life. As a result, prevention has become a key focus in CVD management. Research shows that timely intervention may prevent up to 80% of premature CVD deaths [12]. Nevertheless, the detection of CVD is often delayed due to its asymptomatic development, which makes treatment more challenging. Early identification of high-risk individuals is therefore crucial [13]. Current clinical guidelines increasingly recommend using risk prediction models, mostly based on traditional statistical methods, to guide early and targeted prevention efforts [14].

Advancements in computer science have led to large and complicated datasets which require complex analytical and predictive models. Effective models must manage data complexity while aligning with clinical objectives, generalizing across datasets, remaining robust under varying conditions, and providing consistent, reliable results in real-world settings [15]. This study aimed to identify risk factors associated with CVD using longitudinal data collected over ten years from participants in the Tehran Lipid and Glucose Study (TLGS), a large population-based cohort. We used longitudinal health data to improve the prediction of cardiovascular disease. The data provide an opportunity to employ deep learning models, especially long short-term memory (LSTM) and Gated Recurrent Unit (GRU) networks, which are well suited to analyzing sequential data. The purpose of this research was to assess the performance of deep learning models (LSTM, GRU) and a mixed-effects logistic model as conventional regression approaches by developing and validating them on longitudinal TLGS data to predict 10-year incident CVD.

## Methods

### Participants

TLGS is a large-scale, population-based cohort study initiated in Tehran to investigate risk factors for NCDs. The study was conducted in several phases, starting with a cross-sectional sampling of 15,005 individuals aged 3 years and older. In the following phases, longitudinal follow-up assessments were conducted, which led to a mean participation rate of 73% per phase. The study design has been described in previous publications [16].

For the present analysis, data were initially obtained from 6,516 adults aged 30 years or older who participated in Phase III of the TLGS and had available records from Phase I. Of these, 1,644 individuals were excluded based on the following criteria: prevalent CVD at baseline ($n = 815$), loss to follow-up ($n = 29$), having only one measurement during the follow-up period ($n = 404$), only one measurement before the occurrence of CVD ($n = 139$), or two consecutive missing observations before a CVD event ($n = 257$). Of the 4,872 participants in the final analytic sample, 1,942 were men and 2,930 were women without a previous history of CVD (stroke or CHD). Participants had no prior history of coronary artery disease (including angina, heart attack, coronary artery bypass surgery, or angioplasty), cerebrovascular events (such as stroke or transient ischemic attacks), or peripheral arterial disease (such as claudication). Follow-up continued through March 2020 (Fig. 1). As a large-scale, population-based study, it remains an important resource for tracking risk factors linked to chronic and cardiovascular diseases.

### Measurements

At baseline, participants completed interviews on sociodemographic, medical, and lifestyle factors, including education, smoking, physical activity (assessed by MAQ), employment, marital status, and family history of CVD/diabetes. Anthropometric measures (weight, height, BMI, waist circumference) and blood pressure were obtained by trained staff using standardized procedures.

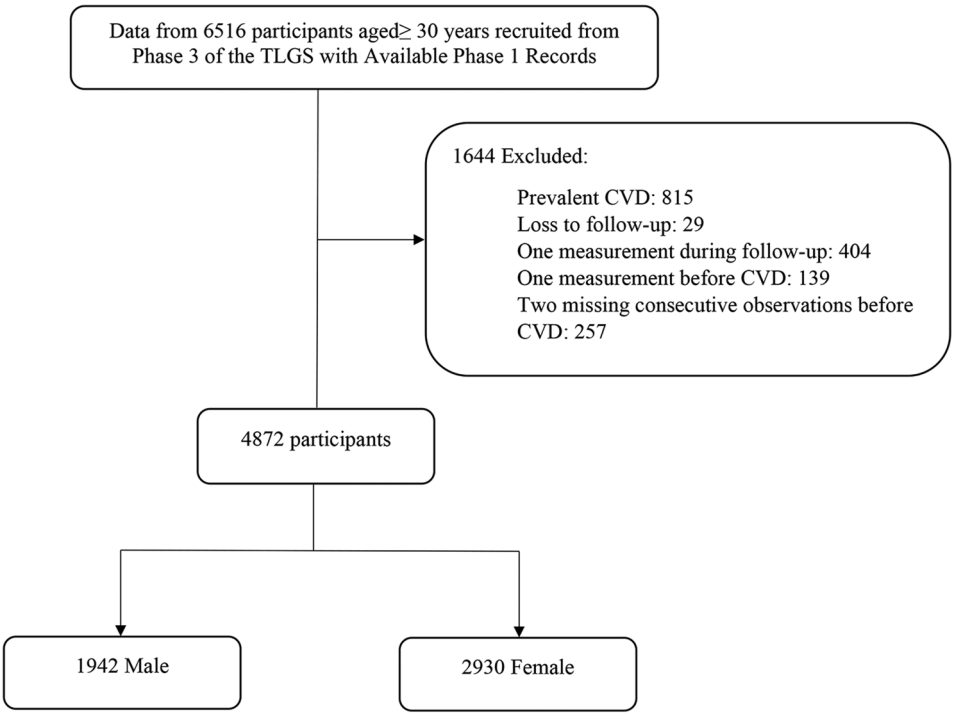



**Fig. 1** Study flowchart illustrating the selection of participants from the Tehran Lipid and Glucose Study (TLGS) cohort

Fasting blood samples were analyzed for glucose, lipid profile, HDL-C, and serum creatinine with validated laboratory methods and quality controls. For full methodological details and interviews used in our study, refer to earlier TLGS publications [17].

#### Definition of outcomes

Participants were followed annually for CVD events via nurse-led phone interviews. Suspected cases were verified by physicians through home visits or hospital record reviews, and diagnoses were confirmed by a panel. According to predefined criteria, CVD was defined as the first occurrence of either coronary heart disease (CHD) or stroke.

#### Analysis method

Variables were reported as mean ± standard deviation (SD) for normally distributed data, median (25th-75th percentile) for skewed data, and as frequencies (percentages) for categorical variables. The differences in normally distributed continuous variables, skewed continuous variables, and categorical variables were assessed through independent t-tests, Mann–Whitney U tests, and chi-squared tests, respectively. These tests were used to compare variables between participants with and without CVD. The date of CVD diagnosis was set as the event date. If participants moved away from the study area, died from non-CVD causes, were lost to follow-up, or reached the end of the follow-up period without having experienced a CVD event, they were censored.

For individuals who developed CVD, the event date marked time zero. For those who did not, the date of the final visit was used as time zero. Data were retrospectively examined in 3-year intervals (at 0, 3, 6, and 9 years prior to the event or censoring) to evaluate the progression of risk factors. The multiple imputation by chained equations (MICE) method with $M = 1$ was utilized to address missing data for covariates at each phase. Linear regression models were applied for continuous variables and logistic regression for binary variables.

The cohort's dataset was used in this study from Phase I through Phase VI. Whenever a CVD event occurred, that time point was designated as time zero, and data from the preceding 10 years were analyzed retrospectively. As a result, the follow-up duration was flexible (e.g., phases III–VI, II–V, or I–IV), to ensure all individuals had a 10-year observation period prior to the outcome. The longitudinal monitoring of risk factors before the incidence of CVD was facilitated by this retrospective design. Machine learning methods, specifically RNNs, were used to model time-dependent relationships among variables in a longitudinal dataset to predict CVD incidence, despite incomplete data. Missing values were imputed prior to modeling in order to address the challenge caused by incomplete data. This preprocessing ensured that all variables included in the RNN analyses had complete data.

The TLGS dataset was used to classify CVD events using 21 features, and two deep learning architectures, LSTM and GRU, were utilized. The training and evaluation of the model were performed using a stratified data split. In particular, 20% of the dataset was selected as a random test set, whereas the remaining 80% was divided into training (64%) and validation (16%) subsets. Stratified sampling by participant maintained the original class distribution across all subsets and eliminated leakage across phases. The training set assisted model parameter optimization, the validation set provided hyperparameter tuning and performance assessment, and the test set was only used for the final evaluation of the model's generalization capability. The LSTM model was set up with a learning rate of 0.01, a dropout rate of 50%, and a mini-batch size of 64. The selection of features for male and female participants was driven by clinical expert opinion and literature, considering sex-specific differences in CVD risk factors and incidence [18].

RNN models were designed to detect patterns over the 10-year follow-up period. Since accuracy may be misleading when faced with unbalanced data, the AUC was used to evaluate model performance. The AUC was selected as the most reliable metric for comparing model performance in terms of risk stratification. Permutation importance was determined by sequentially shuffling each (feature, phase) column across participants, recalculating predictions, and reporting the AUC reduction compared to the un-permuted baseline (negative values indicate noise rather than actual meaning).

Mixed-effects logistic regression with repeated measurements per participant was used to predict first cardiovascular disease events as a traditional model. Model performance was evaluated using a nonparametric bootstrap (2,000 resamples with replacement) to derive a 95% confidence interval (CI) for the ROC area under the curve (AUC). All analyses were conducted in Python and R, and a two-tailed $p$-value $< 0.05$ was considered statistically significant.

## Results

A total of 1942 men and 2930 women were followed up. During a 10-year follow-up, out of 4,872 non-CVD participants in the TLGS, 545 (11.2%) developed CVD. The incidence of CVD was 14.4% (280 of 1,942) in men and 9.0% (265 of 2,930) in women. Table 1 (men) and Table 2 (women) present baseline characteristics at 9, 6, and 3 years before CVD diagnosis and at the time of CVD onset, stratified by CVD status. In both sexes, participants who developed CVD were consistently older and had higher levels of non-HDL cholesterol, triglycerides, BMI, waist circumference, blood pressure, fasting plasma glucose, and 2-hour post-load glucose compared with non-CVD participants across all time points (all $P < 0.05$). HDL cholesterol levels were generally lower in the CVD group, particularly at earlier time points. Estimated GFR was significantly lower among CVD participants, with a progressive decline approaching the CVD event. Lifestyle and socio-demographic differences were also observed. In both men and women, those with CVD had a lower prevalence of being married, employed, and physically active, and a higher prevalence of low educational attainment (≤ 12 years). The proportions of participants using antihypertensive, lipid-lowering, and diabetes medications were substantially higher in the CVD group and increased as the CVD event approached. No significant differences were observed in current smoking rates or family history of CVD and diabetes between groups in either sex.

Two deep learning models, LSTM and GRU, were evaluated in Table 3 for their capacity to predict a health-related outcome based on clinical, biochemical, and demographic characteristics. Each model's performance on the test and validation datasets was presented and stratified by sex. The LSTM model achieved 83% accuracy and AUC = 0.717 on the test dataset and AUC = 0.726 on the validation dataset for men. For women, the LSTM demonstrated AUC = 0.729 and 77% accuracy on the test dataset (validation: AUC = 0.725, 75% accuracy). The GRU achieved higher AUCs than LSTM in all groups—men: 0.735 (test), 0.746 (validation); women: 0.731 (test), 0.727 (validation)—and higher accuracy for women (85% on both sets) but lower accuracy for men (79% test; 78% validation). Recall and precision showed mixed patterns: in men, GRU improved recall relative to LSTM but offered similar or slightly lower precision on validation, whereas in women, GRU yielded higher precision but lower recall.

Permutation importance (AUC reduction after the shuffling of a single feature at a particular phase) indicated that discrimination was mostly focused on late-phase metabolic and blood pressure indicators. The strongest impacts were seen for FPG, phase IV (≈ 0.018), SBP, phase IV (≈ 0.013), non-HDL, phase I (≈ 0.015), 2hPG, phase III (≈ 0.010), and TG, phase IV (≈ 0.011). Most other cells were near zero, suggesting little unique contribution; slight negative values probably indicate sampling noise or minor over fitting (Fig. 2).

Discrimination was primarily affected by late-phase glycemic control and blood pressure, but baseline obesity provided earlier insights. The most significant impacts were 2hPG, phase IV (≈ 0.031), SBP, phase IV (≈ 0.024), and BMI, phase I (≈ 0.024), with supplementary contributions from non-HDL (phase IV ≈ 0.016, phase I ≈ 0.012), TG, phase IV (≈ 0.015), and FPG, phase IV (≈ 0.013). Socio-demographic and lifestyle factors were mostly negligible; minor early-phase negative values for eGFR likely suggest redundancy rather than inverse effects (Fig. 3).

**Table 1** Baseline characteristics of men's participants

| Variables | | 9 year before CVD (*n* = 1942) | 6 year before CVD (*n* = 1942) | 3 year before CVD (*n* = 1942) | CVD (*n* = 1942) |
|---|---|---|---|---|---|
| Non-HDL (mg/dL) | CVD | 169.23 ± 38.01 | 160.16 ± 35.36 | 158.17 ± 39.32 | 150.23 ± 38.55 |
| | Non-CVD | 153.9 ± 35.91 | 150.68 ± 38.43 | 150.88 ± 37.05 | 145.93 ± 38.67 |
| | P-value | < 0.001 | < 0.001 | 0.003 | 0.085 |
| HDL-C (mg/dL) | CVD | 37.15 ± 9.4 | 38.46 ± 9.35 | 41.29 ± 9.77 | 42.47 ± 9.09 |
| | Non-CVD | 37.64 ± 8.47 | 42.94 ± 9.94 | 44.26 ± 10.37 | 43.55 ± 10.22 |
| | P-value | 0.383 | < 0.001 | < 0.001 | 0.073 |
| TG (mg/dL) † | CVD | 166.48(115.75,228) | 152.5(109,207.75) | 153.5(110,197.75) | 143(107.25,196.75) |
| | Non-CVD | 149(105,215) | 146(103,206) | 148(103,207) | 140.58(97,199) |
| | P-value | 0.012 | 0.373 | 0.509 | 0.495 |
| BMI (kg/m$^2$) | CVD | 27.54 ± 4.1 | 27.45 ± 4.09 | 27.68 ± 4.06 | 27.79 ± 4.27 |
| | Non-CVD | 27.14 ± 4.08 | 27.43 ± 4.22 | 27.58 ± 4.36 | 27.87 ± 4.53 |
| | P-value | 0.133 | 0.941 | 0.724 | 0.774 |
| Wc (cm) | CVD | 95.24 ± 11.08 | 97.42 ± 10.71 | 98.72 ± 10.76 | 98.46 ± 10.89 |
| | Non-CVD | 96.11 ± 10.18 | 97.5 ± 10.83 | 97.77 ± 10.74 | 97.11 ± 11.15 |
| | P-value | 0.221 | 0.905 | 0.171 | 0.060 |
| eGFR (mL/min/1.73 m$^2$) | CVD | 74.06 ± 13.56 | 72.45 ± 13.11 | 67.81 ± 12.17 | 64.42 ± 13.17 |
| | Non-CVD | 76.76 ± 13.01 | 75.45 ± 12.83 | 70.64 ± 13.28 | 68.01 ± 13.14 |
| | P-value | 0.001 | < 0.001 | 0.001 | < 0.001 |
| SBP (mmHg) | CVD | 127.15 ± 21.08 | 126.99 ± 19.8 | 130.7 ± 20.34 | 129.94 ± 19.65 |
| | Non-CVD | 117.63 ± 16.52 | 119.82 ± 16.75 | 121.91 ± 17.86 | 121.99 ± 17.66 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| DBP (mmHg) | CVD | 79.63 ± 11.66 | 79.89 ± 11.42 | 81.38 ± 11.33 | 81.65 ± 11.08 |
| | Non-CVD | 76.16 ± 9.67 | 79.71 ± 10.6 | 80.04 ± 10.36 | 79.31 ± 10.33 |
| | P-value | < 0.001 | 0.795 | 0.066 | 0.001 |
| FPG (mg/dL) | CVD | 103.06 ± 34.11 | 107.1 ± 33.76 | 114.36 ± 45.23 | 113.79 ± 44.23 |
| | Non-CVD | 96.7 ± 25.96 | 103.48 ± 27.1 | 105.44 ± 30.55 | 105.21 ± 33.53 |
| | P-value | 0.003 | 0.089 | 0.002 | 0.002 |
| 2hPG (mg/dL) | CVD | 138.68 ± 84.81 | 138.53 ± 78.05 | 156.73 ± 96.57 | 157.13 ± 89.2 |
| | Non-CVD | 117.22 ± 62.38 | 123.36 ± 65.34 | 134.46 ± 72.57 | 144.71 ± 70.43 |
| | P-value | < 0.001 | 0.002 | < 0.001 | 0.027 |
| Age, year | CVD | 53.54 ± 11.98 | 56.85 ± 11.92 | 60.07 ± 11.86 | 63.31 ± 11.8 |
| | Non-CVD | 47.79 ± 12.48 | 51.13 ± 12.48 | 54.34 ± 12.45 | 57.64 ± 12.48 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| FH-DM, n(%) | CVD | 97(34.6) | 107(38.2) | 108(38.6) | 114(40.7) |
| | Non-CVD | 632(38.0) | 684(41.2) | 706(42.5) | 788(47.4) |
| | P-value | 0.279 | 0.354 | 0.220 | 0.038 |
| Spouse, married | CVD | 271(96.8) | 269(96.1) | 267(95.4) | 265(94.6) |
| | Non-CVD | 1552(93.4) | 1533(92.2) | 1557(93.7) | 1548(93.1) |
| | P-value | 0.028 | 0.022 | 0.278 | 0.350 |
| Occupation, employed | CVD | 177(63.2) | 164(58.6) | 143(51.1) | 123(43.9) |
| | Non-CVD | 1302(78.3) | 1195(71.9) | 1104(66.4) | 995(59.9) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| FH-CVD | CVD | 75(26.8) | 77(27.5) | 82(29.3) | 83(29.6) |
| | Non-CVD | 336(20.2) | 363(21.8) | 381(22.9) | 406(24.4) |
| | P-value | 0.013 | 0.036 | 0.021 | 0.063 |
| Current smoker | CVD | 55(19.6) | 61(21.8) | 56(20.0) | 50(17.9) |
| | Non-CVD | 401(24.1) | 346(20.8) | 350(21.1) | 326(19.6) |
| | P-value | 0.101 | 0.713 | 0.687 | 0.491 |
| Education, ≤ 12 years | CVD | 240(85.7) | 238(85.0) | 238(85.0) | 232(82.9) |
| | Non-CVD | 1248(75.1) | 1236(74.4) | 1212(72.9) | 1208(72.7) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| Low PA | CVD | 144(51.4) | 118(42.1) | 100(35.7) | 100(35.7) |

**Table 1** (continued)

| Variables | | 9 year before CVD (*n* = 1942) | 6 year before CVD (*n* = 1942) | 3 year before CVD (*n* = 1942) | CVD (*n* = 1942) |
|---|---|---|---|---|---|
| | Non-CVD | 662(39.8) | 683(41.1) | 586(35.3) | 642(38.6) |
| | P-value | < 0.001 | 0.742 | 0.883 | 0.353 |
| DM medication | CVD | 21(7.5) | 26(9.3) | 41(14.6) | 41(14.6) |
| | Non-CVD | 52(3.1) | 99(6.0) | 146(8.8) | 212(12.8) |
| | P-value | < 0.001 | 0.036 | 0.002 | 0.385 |
| HTN medication | CVD | 12(4.3) | 34(12.1) | 54(19.3) | 74(26.4) |
| | Non-CVD | 37(2.2) | 116(7.0) | 187(11.3) | 284(17.1) |
| | P-value | 0.042 | 0.003 | < 0.001 | < 0.001 |
| Lipid medication | CVD | 9(3.2) | 11(3.9) | 30(10.7) | 39(13.9) |
| | Non-CVD | 38(2.3) | 87(5.2) | 133(8.0) | 225(13.5) |
| | P-value | 0.350 | 0.356 | 0.130 | 0.860 |

Abbreviations: HDL, high-density lipoprotein; TG, triglyceride; BMI, body mass index; Wc, waist circumference; eGFR, estimated glomerular filtration rate; SBP, systolic blood pressure; DBP, diastolic blood pressure; FPG, fasting plasma glucose; 2hPG, 2-hour plasma glucose (post 75 g glucose); FH-DM, family history of diabetes mellitus; FH-CVD, family history of CVD; PA, physical activity; DM, diabetes mellitus; HTN, hypertension

Data are given as the mean ± SD or median (IQ 25–75) for continuous variables unless otherwise indicated (†), and data are given as the n(%) for categorical variables

**Table 2** Baseline characteristics of women's participants

| Variables | | 9 year before CVD (*n* = 2930) | 6 year before CVD (*n* = 2930) | 3 year before CVD (*n* = 2930) | CVD (*n* = 2930) |
|---|---|---|---|---|---|
| Non-HDL (mg/dL) | CVD | 184.66 ± 49.18 | 174.67 ± 41.52 | 165.71 ± 45.53 | 155.37 ± 47.89 |
| | Non-CVD | 153.87 ± 38.92 | 148.71 ± 39.37 | 148.13 ± 38.44 | 146.18 ± 39.09 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | 0.003 |
| HDL-C (mg/dL) | CVD | 42.72 ± 10.36 | 44.76 ± 11.41 | 48.11 ± 11.88 | 49.62 ± 10.43 |
| | Non-CVD | 44.36 ± 10.45 | 50.32 ± 11.2 | 52.29 ± 12.38 | 50.74 ± 11.1 |
| | P-value | 0.015 | < 0.001 | < 0.001 | 0.115 |
| TG (mg/dL) † | CVD | 177(127.66,248) | 175(124.5,240) | 157(117,210) | 154(119.5,200) |
| | Non-CVD | 132(94,190) | 128(91,183.88) | 130(94,184.53) | 129(94,178) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| BMI (kg/m$^2$) | CVD | 30.1 ± 4.7 | 30.77 ± 5.18 | 30.63 ± 4.94 | 30.87 ± 4.96 |
| | Non-CVD | 28.98 ± 4.86 | 29.63 ± 5.04 | 29.88 ± 5.08 | 29.82 ± 5.11 |
| | P-value | < 0.001 | < 0.001 | 0.022 | 0.001 |
| Wc (cm) | CVD | 97.08 ± 11.17 | 98.96 ± 11.28 | 99.4 ± 10.86 | 101.2 ± 10.28 |
| | Non-CVD | 90.9 ± 12.38 | 95.48 ± 11.94 | 95.67 ± 11.94 | 97.59 ± 11.97 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| eGFR (mL/min/1.73 m$^2$) | CVD | 67.04 ± 12.16 | 65.52 ± 12.62 | 62.19 ± 12.32 | 57.71 ± 12.88 |
| | Non-CVD | 72.87 ± 13.86 | 70.65 ± 12.89 | 67.33 ± 13.06 | 62.96 ± 12.49 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| SBP (mmHg) | CVD | 128.74 ± 21.79 | 129.4 ± 20.99 | 128.93 ± 21.44 | 132.29 ± 21.43 |
| | Non-CVD | 113.14 ± 17.81 | 116.97 ± 18.85 | 119.36 ± 19.53 | 117.57 ± 18.33 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| DBP (mmHg) | CVD | 79.92 ± 11.33 | 78.85 ± 11.35 | 79.07 ± 11.52 | 79.19 ± 12.16 |
| | Non-CVD | 73.08 ± 9.82 | 76.38 ± 10.54 | 77.62 ± 10.34 | 76.38 ± 9.79 |
| | P-value | < 0.001 | < 0.001 | 0.032 | < 0.001 |
| FPG (mg/dL) | CVD | 118.43 ± 52.56 | 118.21 ± 48.46 | 123.34 ± 54.51 | 122.75 ± 53.95 |
| | Non-CVD | 95.49 ± 26.36 | 102.04 ± 30.15 | 103.31 ± 31.61 | 102.27 ± 31.21 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| 2hPG (mg/dL) | CVD | 180.14 ± 117.58 | 178.14 ± 106.52 | 182.64 ± 114.28 | 182.54 ± 111.6 |
| | Non-CVD | 121.67 ± 61.91 | 130.0 ± 65.96 | 137.89 ± 71.95 | 142.61 ± 64.83 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| Age, year | CVD | 56.04 ± 10.39 | 59.29 ± 10.39 | 62.48 ± 10.41 | 65.82 ± 10.38 |
| | Non-CVD | 46.45 ± 11.12 | 49.78 ± 11.13 | 52.98 ± 11.15 | 56.29 ± 11.13 |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |

**Table 2** (continued)

| Variables | | 9 year before CVD (*n* = 2930) | 6 year before CVD (*n* = 2930) | 3 year before CVD (*n* = 2930) | CVD (*n* = 2930) |
|---|---|---|---|---|---|
| FH-DM, n(%) | CVD | 108(40.8) | 113(42.6) | 121(45.7) | 131(49.4) |
| | Non-CVD | 1077(40.4) | 1172(44.0) | 1276(47.9) | 1386(52.0) |
| | P-value | 0.914 | 0.676 | 0.490 | 0.424 |
| Spouse, married | CVD | 199(75.1) | 174(65.7) | 174(65.7) | 169(63.8) |
| | Non-CVD | 2279(85.5) | 2202(82.6) | 2163(81.2) | 2073(77.8) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| Occupation, employed | CVD | 15(5.7) | 13(4.9) | 7(2.6) | 6(2.3) |
| | Non-CVD | 341(12.8) | 356(13.4) | 336(12.6) | 366(13.7) |
| | P-value | 0.001 | < 0.001 | < 0.001 | < 0.001 |
| FH-CVD | CVD | 62(23.4) | 72(27.2) | 81(30.6) | 87(32.8) |
| | Non-CVD | 621(23.3) | 691(25.9) | 736(27.6) | 786(29.5) |
| | P-value | 0.972 | 0.661 | 0.307 | 0.257 |
| Current smoker | CVD | 8(3.0) | 5(1.9) | 7(2.6) | 6(2.3) |
| | Non-CVD | 76(2.9) | 86(3.2) | 89(3.3) | 94(3.5) |
| | P-value | 0.876 | 0.230 | 0.543 | 0.280 |
| Education, ≤ 12 years | CVD | 256(96.6) | 255(96.2) | 257(97.0) | 257(97.0) |
| | Non-CVD | 2314(86.8) | 2726(85.9) | 2239(84.0) | 2211(83.0) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| Low PA | CVD | 123(46.4) | 102(38.5) | 89(33.6) | 109(41.1) |
| | Non-CVD | 836(31.4) | 677(25.4) | 770(28.9) | 822(30.8) |
| | P-value | < 0.001 | < 0.001 | 0.110 | 0.001 |
| DM medication | CVD | 45(17.0) | 58(21.9) | 70(26.4) | 89(33.6) |
| | Non-CVD | 121(4.5) | 196(7.4) | 294(11.0) | 400(15.0) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| HTN medication | CVD | 53(20.0) | 62(23.4) | 99(37.4) | 134(50.6) |
| | Non-CVD | 142(5.3) | 358(13.4) | 516(19.4) | 661(24.8) |
| | P-value | < 0.001 | < 0.001 | < 0.001 | < 0.001 |
| Lipid medication | CVD | 22(8.3) | 37(14.0) | 65(24.5) | 95(35.8) |
| | Non-CVD | 141(5.3) | 242(9.1) | 415(15.6) | 627(23.5) |
| | P-value | 0.041 | 0.010 | < 0.001 | < 0.001 |

Abbreviations: HDL, high-density lipoprotein; TG, triglyceride; BMI, body mass index; Wc, waist circumference; eGFR, estimated glomerular filtration rate; SBP, systolic blood pressure; DBP, diastolic blood pressure; FPG, fasting plasma glucose; 2hPG, 2-hour plasma glucose (post 75 g glucose); FH-DM, family history of diabetes mellitus; FH-CVD, family history of CVD; PA, physical activity; DM, diabetes mellitus; HTN, hypertension

Data are given as the mean ± SD or median (IQ 25–75) for continuous variables unless otherwise indicated (†), and data are given as the n(%) for categorical variables

Traditional trend analyses of longitudinal data can reveal differences between participants who develop CVD ("converters") and those who do not ("non-converters"); however, because these approaches evaluate predictors one at a time, they do not capture higher-order interactions. Sex-stratified mixed-effects logistic regression with repeated measurements per participant was used to predict first cardiovascular disease events as a traditional comparison. Discrimination was defined by the area under the ROC curve (AUC) with 95% confidence intervals; performance was determined on the held-out test dataset using marginal predicted probabilities. On the test dataset, the mixed-effects model obtained an AUC of 0.671 (0.635, 0.707) in men and 0.759 (0.726, 0.790) in women, respectively. Even slight enhancements in discrimination may be significant at the population level, particularly for preventive screening, where small improvements may result in the early identification of thousands more at-risk participants.

Compared with traditional mixed-effects models, our GRU-based approach provides time-dependent variable importance via feature–phase permutation-importance heat maps—insight not readily available from conventional models. On the test dataset, discrimination was higher in men for GRU (AUC 0.735; 95% CI 0.683–0.783) than for the mixed-effects model (0.671; 95% CI 0.635–0.707), whereas in women GRU was slightly lower than mixed-effects (0.731; 95% CI 0.679–0.780 vs. 0.759; 95% CI 0.726–0.790), with overlapping confidence intervals. These findings suggest that deep learning can leverage longitudinal health records to capture time-dependent risk patterns and—depending on subgroup—can match or exceed traditional approaches while offering richer temporal interpretability. Nonetheless, external

**Table 3** Model Building strategy and evaluation criteria stratified by sex on test and validation sets

| Models | | The feature considered | Test data | | | | Validation data | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | AUC (%95 CI) | Recall | Accuracy | Precision | AUC (%95 CI) | Recall | Accuracy | Precision |
| LSTM | | | | | | | | | | |
| | Men | Non-HDL, HDL, TG, BMI, Wc, eGFR, SBP, DBP, FPG, 2hPG, Age, FH-DM, Spouse, occupation, FH-CVD, smoking, Education, PA, DM medication, HTN medication, and Lipid medication | 0.717 (0.664,0.766) | 28% | 83% | 31/119(26%) | 0.726 (0.673,0.778) | 32% | 84% | 28/95(29%) |
| | Women | Non-HDL, HDL, TG, BMI, Wc, eGFR, SBP, DBP, FPG, 2hPG, Age, FH-DM, Spouse, occupation, FH-CVD, smoking, Education, PA, DM medication, HTN medication, and Lipid medication | 0.729 (0.678,0.776) | 50% | 77% | 55/230(24%) | 0.725 (0.672,0.779) | 52% | 75% | 45/198(23%) |
| GRU | | | | | | | | | | |
| | Men | Non-HDL, HDL, TG, BMI, Wc, eGFR, SBP, DBP, FPG, 2hPG, Age, FH-DM, Spouse, occupation, FH-CVD, smoking, Education, PA, DM medication, HTN medication, and Lipid medication | 0.735 (0.683,0.783) | 50% | 79% | 54/199(27%) | 0.746 (0.695,0.795) | 47% | 78% | 41/165(25%) |
| | Women | Non-HDL, HDL, TG, BMI, Wc, eGFR, SBP, DBP, FPG, 2hPG, Age, FH-DM, Spouse, occupation, FH-CVD, smoking, Education, PA, DM medication, HTN medication, and Lipid medication | 0.731 (0.679,0.780) | 21% | 85% | 23/85(27%) | 0.727 (0.674,0.782) | 25% | 85% | 22/77(29%) |

Abbreviations: HDL, high-density lipoprotein; TG, triglyceride; BMI, body mass index; Wc, waist circumference; eGFR, estimated glomerular filtration rate; SBP, systolic blood pressure; DBP, diastolic blood pressure; FPG, fasting plasma glucose; 2hPG, 2-hour plasma glucose (post 75 g glucose); FH-DM, family history of diabetes mellitus; FH-CVD, family history of CVD; PA, physical activity; DM, diabetes mellitus; HTN, hypertension

validation is needed to confirm reproducibility and clinical utility.

## Discussion

When utilizing large-scale datasets to identify new risk factors and model complex temporal patterns, data-driven techniques, such as machine learning, may improve disease risk discrimination. Using longitudinal data from the TLGS, this study examined the predictive performance of deep learning models, specifically RNN architectures such as LSTM and GRU, for predicting CVD events over a 10-year follow-up. RNN models were implemented, which are particularly well-suited to handling longitudinal data, thereby facilitating the detection of non-linear relationships and interactions among predictors. Using 21 routinely measured variables, GRU models consistently outperformed LSTM models in both men and women (test AUC: men 0.738 vs. 0.719; women 0.739 vs. 0.723; validation AUCs were similar). In sex-stratified analyses, predictive discrimination was superior to the mixed-effects model in men (AUC 0.738 vs. 0.70) but comparable in women (AUC 0.739 vs. 0.74). In addition to model performance evaluation, verification procedures were conducted to ensure reliability.

Current cardiovascular disease risk prediction methods recommended in clinical guidelines often demonstrate inconsistent performance across various populations. The approach we used included baseline and repeated follow-up measures, establishing a longitudinal framework for prediction. Comparing our findings with other studies reveals that differences in study aims, data formats, and follow-up periods explain the observed performance inconsistencies. Arunachalam (2020) employed six classification algorithms, including SVM and MLP, on the Cleveland dataset for cardiovascular disease detection employing 14 features, with AUCs over 0.90. However, their research focused on the diagnosis of CVD using cross-sectional data, while our study focused on predicting CVD—a more complex task with lower AUC values [19]. Choi et al. (2017) utilized GRU-based RNN models for predicting heart failure within 12–18 months, with AUCs of 0.777 and 0.883 for the shorter and longer observation periods, respectively. Their performance may be related to a large sample size and shorter prediction horizon, which introduces uncertainty relative to our 10-year prediction interval [20].

Sung et al. (2019) created deep learning and Cox regression models to predict cardiovascular disease using data from more than 360,000 participants in the NHIS-HEALS cohort. Their deep learning model reached AUCs greater than 0.94 at the 2-year mark, which is significantly higher than our findings. The difference is likely attributable to the shorter follow-up period and a considerably larger sample size [21]. Alaa et al. (2019)

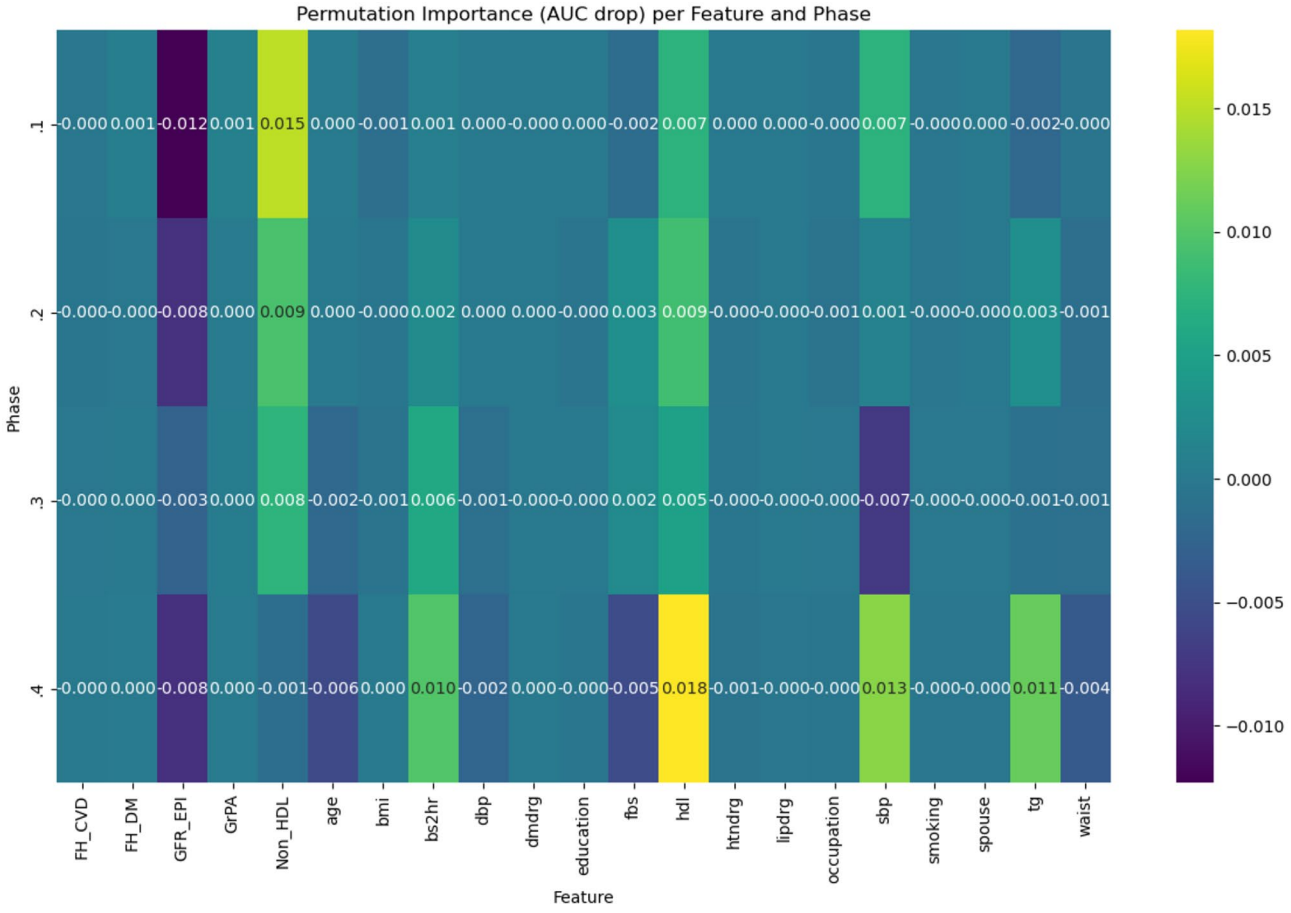



**Fig. 2** The GRU model for women: permutation importance (AUC decrease) per feature and phase. After randomly permuting one feature at one phase across subjects, each cell illustrates the decrease in validation ROC–AUC. The importance of the feature is demonstrated by warmer colors. 1, 2, 3, and 4 represent 9 years before CVD, 6 years before CVD, 3 years before CVD, and the time of CVD, respectively

performed a large-scale study employing the AutoPrognosis ML framework with more than 423,000 participants from the UK Biobank, evaluating 473 variables and obtaining an AUC of 0.774. Despite the wide range of features of their model, our GRU-based model reached a similar AUC (~ 0.74) with only 21 accessible clinical variables. This underscores the cost-effectiveness and applicability of our approach, as models with fewer predictors are more easily implemented in real-world healthcare settings without substantial loss of performance [22].

In general, our results suggest that GRU models could effectively predict long-term CVD risk with reasonable accuracy using longitudinal medical data with restricted features. While larger datasets, shorter prediction horizons, and more extensive variable sets may enhance predictive performance (e.g., higher AUCs), our findings demonstrate that efficient, low-cost, and clinically implementable deep learning models are feasible for CVD risk prediction in population-based environments. However, the internal validation procedures performed in the present research (stratified sampling, held-out test data, and permutation importance analyses) show a preliminary sign of the model's stability. Nevertheless, these are insufficient to replace external validation, which is essential prior to clinical application.

## Conclusions

The findings emphasize the potential of deep learning approaches to improve predictive discrimination by leveraging longitudinal medical data and identifying time-dependent patterns in risk factors, compared with traditional methods. Although larger sample sizes and more comprehensive features can improve model performance, our research shows the cost-effectiveness and feasibility of models constructed using a limited number of routinely available medical characteristics without losing predictive power. Additional data types (e.g., imaging, genetic, and lifestyle data) should be included in these models, and explainable AI techniques should be investigated to improve clinical interpretability. Future research should focus on validating these models in other cohorts. Finally, deep learning models could be helpful in the

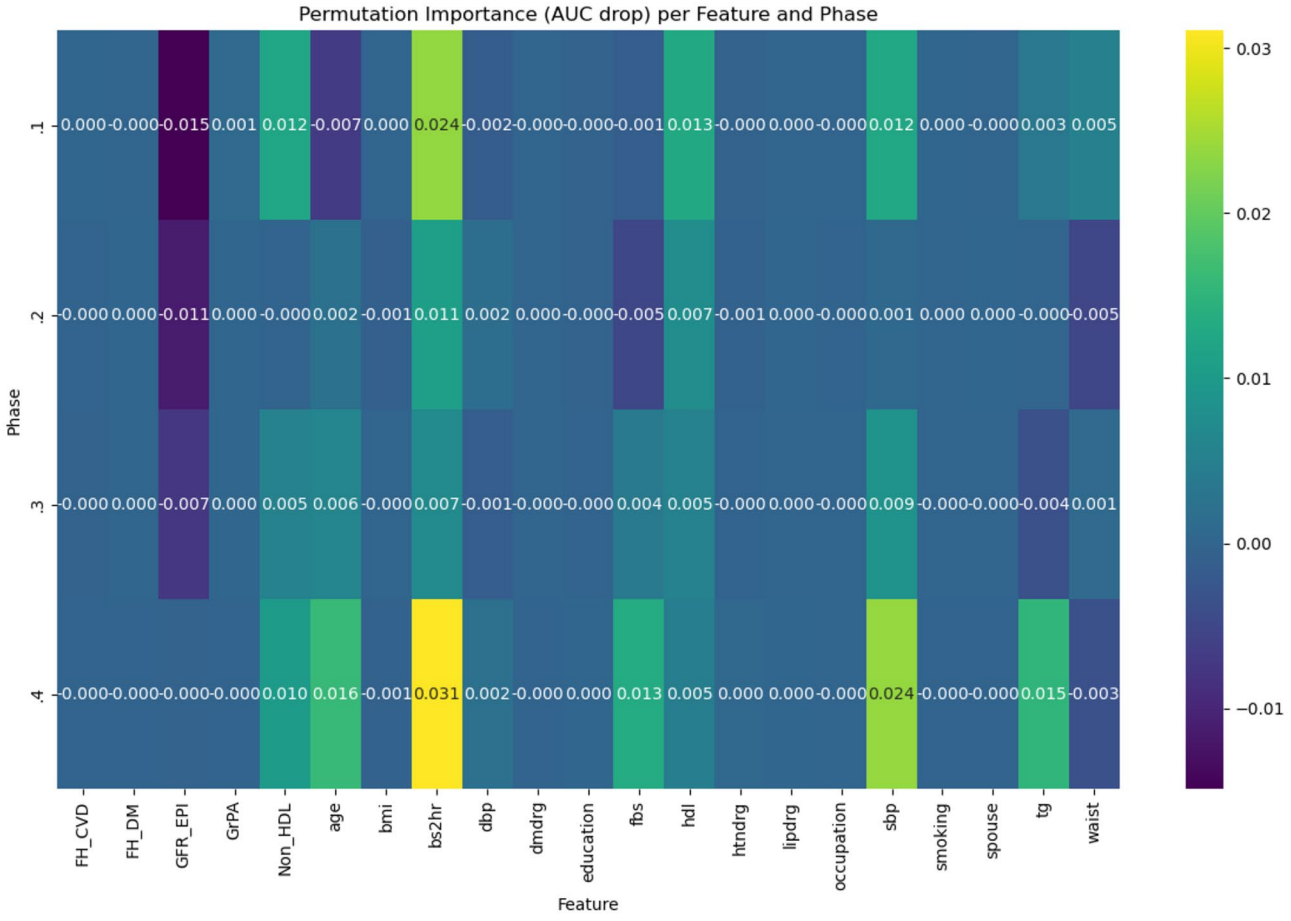



**Fig. 3** The GRU model for men: permutation importance (AUC decrease) per feature and phase. After randomly permuting one feature at one phase across subjects, each cell illustrates the decrease in validation ROC–AUC. The importance of the feature is demonstrated by warmer colors. 1, 2, 3, and 4 represent 9 years before CVD, 6 years before CVD, 3 years before CVD, and the time of CVD, respectively

early classification of CVD risk to implement preventive interventions.

**Abbreviations**

| | |
|---|---|
| CVD | Cardiovascular disease |
| TLGS | Tehran Lipid and Glucose Study |
| NCDs | Non-communicable diseases |
| DALY | Disability-adjusted life year |
| MAQ | Modifiable Activity Questionnaire |
| SD | Standard deviation |
| RNN | Recurrent neural network |
| CHD | Coronary heart disease |
| LSTM | Long Short-Term Memory |
| GRU | Gated Recurrent Unit |
| AUC | Area under the receiver operating characteristic curve |
| HDL-C | High-Density Lipoprotein-Cholesterol |
| TG | Triglyceride |
| BMI | Body Mass Index |
| Wc | Waist circumference |
| eGFR | estimated Glomerular filtration rate |
| SBP | Systolic Blood Pressure |
| DBP | Diastolic Blood Pressure |
| FPG | Fasting plasma glucose |
| 2hPG | 2 h Post 75 g Glucose |
| FH-DM | Family history of diabetes mellitus |
| FH-CVD | Family history of CVD |
| PA | Physical activity |
| DM | Diabetes mellitus |
| HTN | Hypertension |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12911-025-03280-5.

Supplementary Material 1

Supplementary Material 2

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**Author contributions**

'MM': Data collection, Study design, literature review, and manuscript preparation. 'DK': Study design, revising the manuscript, and final approval of the manuscript. 'MM', 'AT', and 'KH': Data analysis and data interpretation. 'AA' and 'AK': revising the manuscript, and the final approval of the manuscript. The final draft was evaluated and approved by all authors.

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**Data availability**
The datasets analyzed during this study are available from the corresponding author upon reasonable request.

**Code availability**
All source code required to reproduce the analyses is publicly available: GRU and LSTM implementations (Supplementary File S1) and mixed-effects model code (Supplementary File S2).

## Declarations

**Ethics approval and consent to participate**
The data used in this study were obtained from the Tehran Lipid and Glucose Study (TLGS), carried out by the Endocrine Research Center at Shahid Beheshti University of Medical Sciences. The authors express their appreciation to all participants and individuals involved in the design and data collection of the TLGS. This project received ethical approval from the Tarbiat Modares University Ethics Committee under the code IR.MODARES.REC.1403.100. This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. All procedures involving human participants received approval from the Human Research Review Committee of the Endocrine Research Center at Shahid Beheshti University in Tehran, Iran. Written informed consent was obtained from all participants.

**Consent for publication**
Not applicable.

**Competing interests**
The authors declare no competing interests.

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**References**

1. Taheri Soodejani M. Non-communicable diseases in the world over the past century: a secondary data analysis. Front Public Health. 2024;12:1436236.
2. Roth GA, Mensah GA, Johnson CO, Addolorato G, Ammirati E, Baddour LM, et al. Global burden of cardiovascular diseases and risk factors, 1990–2019: update from the GBD 2019 study. J Am Coll Cardiol. 2020;76(25):2982–3021.
3. Aminorroaya A, Yoosefi M, Rezaei N, Shabani M, Mohammadi E, Fattahi N, et al. Global, regional, and National quality of care of ischaemic heart disease from 1990 to 2017: a systematic analysis for the global burden of disease study 2017. Eur J Prev Cardiol. 2022;29(2):371–9.
4. Organization WH. Noncommunicable diseases country profiles. 2018 [acceso: 30/07/2019].
5. Rahmani A, Sayehmiri K, Asadollahi K, Sarokhani D, Islami F, Sarokhani M. Investigation of the prevalence of obesity in Iran: a systematic review and meta-analysis study. Acta Medica Iranica. 2015;596–607.
6. Uthman OA. Global, regional, and National disability-adjusted life years (DALYs) for 315 diseases and injuries and healthy life expectancy (HALE) for 195 countries and territories, 1990-2015: a systematic analysis for the global burden of Diseases, Injuries, and risk factors (GBD) 2015 study. Lancet. 2016;388(10053):1603–58.
7. Roth GA, Johnson C, Abajobir A, Abd-Allah F, Abera SF, Abyu G, et al. Global, regional, and National burden of cardiovascular diseases for 10 causes, 1990 to 2015. J Am Coll Cardiol. 2017;70(1):1–25.
8. Danaei G, Farzadfar F, Kelishadi R, Rashidian A, Rouhani OM, Ahmadnia S, et al. Iran in transition. Lancet. 2019;393(10184):1984–2005.
9. Mensah GA, Brown DW. An overview of cardiovascular disease burden in the united States. Health Aff. 2007;26(1):38–48.
10. Jabir R, Nasir Siddiqui N, Kandy Firoz A, Md Ashraf C, Kashif Zaidi G, Shahnawaz Khan S. Current updates on therapeutic advances in the management of cardiovascular diseases. Curr Pharm Design. 2016;22(5):566–71.
11. Guidry UC, Evans JC, Larson MG, Wilson PW, Murabito JM, Levy D. Temporal trends in event rates after Q-wave myocardial infarction: the Framingham heart study. Circulation. 1999;100(20):2054–9.
12. Piepoli MF, Hoes AW, Agewall S, Albus C, Brotons C, Catapano AL, et al. Guidelines: editor's choice: 2016 European guidelines on cardiovascular disease prevention in clinical practice: the sixth joint task force of the European society of cardiology and other societies on cardiovascular disease prevention in clinical practice (constituted by representatives of 10 societies and by invited experts) developed with the special contribution of the European association for cardiovascular prevention & rehabilitation (EACPR). Eur Heart J. 2016;37(29):2315.
13. Liu S, Li Y, Zeng X, Wang H, Yin P, Wang L, et al. Burden of cardiovascular diseases in China, 1990–2016: findings from the 2016 global burden of disease study. JAMA Cardiol. 2019;4(4):342–52.
14. Goff DC Jr, Lloyd-Jones DM, Bennett G, Coady S, D'agostino RB, Gibbons R, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American college of Cardiology/American heart association task force on practice guidelines. Circulation. 2014;129(25suppl2):S49–73.
15. Faizal ASM, Thevarajah TM, Khor SM, Chang S-W. A review of risk prediction models in cardiovascular disease: conventional approach vs. artificial intelligent approach. Comput Methods Programs Biomed. 2021;207:106190.
16. Azizi F, Madjid M, Rahmani M, Emami H, Mirmiran P, Hadjipour R. Tehran lipid and glucose study (TLGS): rationale and design. Iran J Endocrinol Metabolism. 2000;2(2):77–86.
17. Azizi F, Ghanbarian A, Momenan AA, Hadaegh F, Mirmiran P, Hedayati M, et al. Prevention of non-communicable disease in a population in nutrition transition: Tehran lipid and glucose study phase II. Trials. 2009;10:1–15.
18. Rajendran A, Minhas AS, Kazzi B, Varma B, Choi E, Thakkar A, et al. Sex-specific differences in cardiovascular risk factors and implications for cardiovascular disease prevention in women. Atherosclerosis. 2023;384:117269.
19. Arunachalam S. Cardiovascular disease prediction model using machine learning algorithms. Int J Res Appl Sci Eng Technol. 2020;8:1006–19.
20. Choi E, Schuetz A, Stewart WF, Sun J. Using recurrent neural network models for early detection of heart failure onset. J Am Med Inform Assoc. 2017;24(2):361–70.
21. Sung JM, Cho I-J, Sung D, Kim S, Kim HC, Chae M-H, et al. Development and verification of prediction models for preventing cardiovascular diseases. PLoS ONE. 2019;14(9):e0222809.
22. Alaa AM, Bolton T, Di Angelantonio E, Rudd JH, Van der Schaar M. Cardiovascular disease risk prediction using automated machine learning: A prospective study of 423,604 UK biobank participants. PLoS ONE. 2019;14(5):e0213653.

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